

PREVAILING WINDS



TRADE WINDS

The prevailing winds over the tropical oceans are gentle breezes that blow from the northeast or southeast. They are known as the trade winds, because in the days of sailing ships they were used by ocean traders sailing westward. To cross the Atlantic to America, for example, ships sailed south to the tropical islands off Africa, where they picked up the trade winds that took them toward the west.

WESTERLIES

Sailing ships returned east by heading away from the tropics to the temperate zones, to pick up southwesterly or northwesterly winds that would carry them back across the ocean. These westerlies are generally much stronger than the trade winds, especially in the far south where there are no continents to disrupt the airflow. Between 40°S and 50°S they are so strong that they are known as the Roaring Forties.



POLAR EASTERLIES

Near the poles, the prevailing winds blow south off the polar ice and swerve to the west. These polar easterlies drive floating pack ice westward around the Arctic Ocean, and westward around the coasts of Antarctica. In the Antarctic Weddell Sea, the westward flow swirls north along the shores of the Antarctic Peninsula, carrying the ice into the westerly wind zone where it is driven east again.



▲ CALM ZONES

Regions where the air is rising or sinking have very light winds, and often no wind at all. The zones of rising air near the equator are called the doldrums. These calm zones were once a problem for sailing ships, which could be stuck in them for weeks.

▼ MARITIME CLIMATES

Where prevailing winds blow off oceans onto the land, they carry moist air with them. In Ireland, for example, prevailing westerlies create damp, cool conditions that are ideal for lush grass growth.

